

requiring any smoothness of the data distribution.

Guarantees of ODE flows Within the studies of score-based diffusion models (note that the forward process is always SDE), theoretical findings for the ODE reverse process are relatively fewer. To the best of our knowledge, [18] established the first non-asymptotic polynomial convergence rate where the error bound involves an exponential factor in the flow time; [16] provided the first polynomial-time convergence guarantees for the probability flow ODE implementation with a corrector step. Recently, [49] established bounds for both deterministic (ODE) and non-deterministic (SDE) samplers under certain additional assumptions on learning the score. The analysis is done by directly tracking the density ratio between the law of the diffusion process and that of the generated process in discrete time, leading to various non-asymptotic convergence rates.

Compared to score-based diffusion models, the guarantees of flow models (in both forward and reverse processes) are significantly less developed. We are aware of two recent works: The error bounds for the flow-matching model [3] were proved in [10] and applied to probability flow ODE in score-based diffusion models; for neural ODE models trained by likelihood maximization (the framework in [29]), [53] proved non-parametric statistical convergence rates to learn a distribution from data. Both works used a continuous-time formulation, and the flow models therein are trained end-to-end. Compared to end-to-end training, progressive flow models can offer advantages in training efficiency and accuracy, in addition to other advantages such as smaller model complexity. For the analysis, the formulation of progressive flow models is variational and time-discretized in nature. Theoretical studies of time-discretized ODE flow models in both forward and reverse directions remain rudimentary.

1.2.4 Optimization in Wasserstein space

Continuing the classical literature in optimization and information geometry, several recent works established a convergence guarantee of first-order optimization in probability space in various contexts, leveraging the connection to the Wasserstein gradient flow. These analyses can potentially be leveraged under the theoretical framework of this paper to develop new (progressive) flow models as well as theoretical guarantees of generative models.

Optimization in probability distribution space Convergence and rate analysis for first-order methods for vector-space optimization, primarily gradient descent and stochastic gradient descent — sometimes referred to as the Sample Average Approximation (SAA) approach — for convex and strongly convex problems have been established in the original works [55, 56], and extended in various contexts in subsequent papers. Optimization in the space of probability distributions (which forms a manifold) naturally arises in many learning problems and has become an important field of study in statistics and machine learning. In particular, the seminal work of Amari [5, 6] introduced information geometry emerging from studies of a manifold of probability distributions. It includes convex analysis and its duality as a special but important component; however, the line of work did not develop error analysis or convergence rates for algorithms on the probabilistic manifold. More recently, a Frank-Wolfe procedure in probability space was proposed in [40] motivated by applications in nonparametric estimation and was shown to converge exponentially fast under general mild assumptions on the objective functional.

Wasserstein proximal gradient descent The landmark work [39] showed the solution to the Fokker Planck equation as the gradient flow of the KL divergence under the $\mathcal{W}_2$ -distance. The proof in [39] employed a time discretization of the gradient flow now recognized as the JKO scheme. Making

a connection between Langevin Monte Carlo and Wasserstein gradient flow, [12] proposed a proximal version of the Unadjusted Langevin Algorithm corresponding to a splitting scheme of the discrete Wasserstein GD and derived non-asymptotic convergence analysis. To analyze the convergence of discrete-time $\mathcal{W}_2$ -gradient flow, [63] introduced a Forward-Backward time discretization in the proximal Wasserstein GD and proved convergence guarantees akin to the GD algorithm in Euclidean spaces. We comment on the difference between [63] and our scheme in more detail later in Remark 4.1.

The JKO scheme also inspired recent studies in variational inferences (VI). In the context of Gaussian VI, [45] proposed gradient flow of the KL divergence on the Bures-Wasserstein (BW) space, namely the space of Gaussian distributions on $\mathbb{R}^d$ endowed with the $\mathcal{W}_2$ -distance. The algorithm enjoys the explicit solution of the JKO scheme in the BW space, and convergence of the continuous-time gradient flow was proved. In a follow-up work [23], the forward-backward splitting was adopted in the proximal Wasserstein GD in the BW space, leading to convergence guarantees of the discrete-time GD to first-order stationary solutions. The closed-form solution of the JKO operator only applies to the BW space, while the JKO flow network tries to learn a transport map to solve the JKO scheme in each step, leveraging the expressiveness of neural networks. Theoretically, we consider distributions with finite second moments in this work.

1.3 Notations

Throughout the paper, we consider distributions over $\mathcal{X}$ and the domain $\mathcal{X} = \mathbb{R}^d$. We denote by $\mathcal{P}_2$, meaning $\mathcal{P}_2(\mathbb{R}^d)$, the space of probability distributions on $\mathbb{R}^d$ that has finite second moment. Specifically, for a distribution P , define $M_2(P) := \int_{\mathbb{R}^d} \|x\|^2 dP(x)$. When P has a density (with respect to the Lebesgue measure dx), we also write $M_2(P)$ as $M_2(p)$. Then $\mathcal{P}_2 = \{P \text{ on } \mathbb{R}^d, \text{ s.t., } M_2(P) < \infty\}$. We denote by $\mathcal{P}_2^r$ the distributions in $\mathcal{P}_2$ that have densities, namely $\mathcal{P}_2^r = \{P \in \mathcal{P}_2, P \ll dx\}$. We also say a density $p \in \mathcal{P}_2^r$ when $dP(x) = p(x)dx$ is in $\mathcal{P}_2^r$. In this paper, we consider distributions that have densities in most places. When there is no confusion, we use the density p to stand for both the density and the distribution $dP(x) = p(x)dx$, e.g., we say that a random variable $X \sim p$.

Given a (measurable) map $v : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and P a distribution on $\mathbb{R}^d$, its L^2 norm is denoted as $\|v\|_P := (\int_{\mathbb{R}^d} \|v(x)\|^2 dP(x))^{1/2}$. When P has density p , we also denote it as $\|v\|_p$. For $P \in \mathcal{P}_2$, we denote by $L^2(P)$ (and also by $L^2(p)$ when P has density p) the L^2 space of vector fields, that is, $L^2(P) := \{v : \mathbb{R}^d \rightarrow \mathbb{R}^d, \|v\|_P < \infty\}$. For $u, v \in L^2(P)$, define $\langle u, v \rangle_P := \int_{\mathbb{R}^d} u(x)^T v(x) dP(x)$, which is also denoted as $\langle u, v \rangle_p$ when p is the density. The notation Id stands for the identity map, which is always in $L^2(P)$ for $P \in \mathcal{P}_2$. For $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$, the *pushforward* of a distribution P is denoted as $T_{\#}P$, such that $T_{\#}P(A) = P(T^{-1}(A))$ for any measurable set A . When P has density p and $T_{\#}P$ also has a density, we also denote by $T_{\#}p$ the density of $T_{\#}P$. For two maps $S, T : \mathbb{R}^d \rightarrow \mathbb{R}^d$, $S \circ T$ is the function composition.

2 Preliminaries

2.1 Wasserstein-2 distance and optimal transport

We first review the definitions of the Wasserstein-2 distance and optimal transport (OT) map, which are connected by the Brenier Theorem (see, e.g., [7, Section 6.2.3]).

Given two distributions $\mu, \nu \in \mathcal{P}_2$, the Wasserstein-2 distance $\mathcal{W}_2(\mu, \nu)$ is defined as

$$\mathcal{W}_2^2(\mu, \nu) := \inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^2 d\pi(x, y), \quad (10)$$

where $\Pi(\mu, \nu)$ denotes the family of all joint distributions with μ and ν as marginal distributions. When P and Q are in $\mathcal{P}_2^r$ and have densities p and q respectively, we also denote $\mathcal{W}_2(P, Q)$ as $\mathcal{W}_2(p, q)$. When at least one of μ and ν has density, we have the Brenier Theorem, which allows us to define the optimal transport (OT) map from μ to ν .

Theorem 2.1 (Brenier Theorem). *Let $\mu \in \mathcal{P}_2^r$ and $\nu \in \mathcal{P}_2$. Then*

- (i) *There exists a unique minimizer π of (10), which is characterized by a uniquely determined μ -a.e. map $T_\mu^\nu : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that $\pi = (\text{Id}, T_\mu^\nu)_\# \mu$, where (Id, T_μ^ν) maps (x, y) to $(x, T_\mu^\nu(y))$. Moreover, there exists a convex function $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}$ such that $T_\mu^\nu = \nabla \varphi$ μ -a.e.*
- (ii) *The minimum of (10) equals that of the Monge problem, namely*

$$\mathcal{W}_2^2(\mu, \nu) = \inf_{T: \mathbb{R}^d \rightarrow \mathbb{R}^d, T_\# \mu = \nu} \int \|x - T(x)\|^2 d\mu(x).$$

- (iii) *If moreover $\nu \in \mathcal{P}_2^r$, then we also have the OT map T_ν^μ defined ν -a.e., and $T_\nu^\mu \circ T_\mu^\nu = \text{Id}$ μ -a.e., $T_\mu^\nu \circ T_\nu^\mu = \text{Id}$ ν -a.e.*

In most places in our analysis, we will consider the OT between μ and ν both in $\mathcal{P}_2^r$, and we will frequently use the Brenier Theorem (iii) to obtain the pair of OT maps which are inverse of each other in the a.e. sense.

2.2 Differential and convexity of functionals on $\mathcal{P}_2$

Consider a proper lower semi-continuous functional $\phi : \mathcal{P}_2 \rightarrow (-\infty, \infty]$ and we denote the domain to be $\text{Dom}(\phi) = \{\mu \in \mathcal{P}_2, \phi(\mu) < \infty\}$. The subdifferential of ϕ was defined in the Fréchet sense, see, e.g., Definition 10.1.1 of [7]. We recall the definition of strong subdifferential as below.

Definition 2.2 (Strong subdifferential). *Given $\mu \in \mathcal{P}_2$, a vector field $\xi \in L^2(\mu)$ is a strong (Fréchet) subdifferential of ϕ at μ if for $v \in L^2(\mu)$,*

$$\phi((\text{Id} + v)_\# \mu) - \phi(\mu) \geq \langle \xi, v \rangle_\mu + o(\|v\|_\mu).$$

We denote by $\partial_{\mathcal{W}_2} \phi(\mu)$ the set of strong Fréchet subdifferentials of ϕ at μ (which may be empty).

There can be different ways to introduce convexity of functions on $\mathcal{P}_2$. The most common way is the convexity along geodesics, also known as “displacement convexity.” In our analysis, we technically need the notation of convexity *along generalized geodesics* (a.g.g.), which is stronger than geodesic convexity. In short, displacement convexity is along the geodesic from μ_1 to μ_2 , which, in the simple case where there is a unique OT map T_1^2 from μ_1 to μ_2 , is defined using interpolation $(1-t)\text{Id} + tT_1^2$ for $t \in [0, 1]$. In contrast, convexity a.g.g. involves a third distribution ν and is defined using interpolation of the two OT maps from ν to μ_1 and μ_2 respectively.

Specifically, let $\nu \in \mathcal{P}_2^r$, $\mu_i \in \mathcal{P}_2$, $i = 1, 2$, and let T_ν^i be the OT map from ν to μ_i respectively. A *general geodesic* joining μ_1 to μ_2 (with base ν) is a curve of type

$$\mu_t^{1 \rightarrow 2} := ((1-t)T_\nu^1 + tT_\nu^2)_\# \nu, \quad t \in [0, 1]. \quad (11)$$

Definition 2.3 (Convexity along generalized geodesics). *For $\lambda \geq 0$, a functional ϕ on $\mathcal{P}_2$ is said to be λ -convex along generalized geodesics (a.g.g.) if for any $\nu \in \mathcal{P}_2^r$ and $\mu_1, \mu_2 \in \mathcal{P}_2$ and $\forall t \in [0, 1]$,*

$$\phi(\mu_t^{1 \rightarrow 2}) \leq (1-t)\phi(\mu_1) + t\phi(\mu_2) - \frac{\lambda}{2}t(1-t)\mathcal{W}_\nu^2(\mu_1, \mu_2), \quad (12)$$